

Late February 2025

The HUMAN Digest 1.8

Lake Forest College's AI newsletter, courtesy of the HUMAN grant



Image created in Midjourney by David Sanchez-Burr.

Hello.

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“Reasoning” by AI models is all the rage. DeepSeek, based in China, recently released a [reasoning model](#) hot on the heels of OpenAI’s o series of reasoning models, o1 and o3. The widespread notion that the model only cost approximately five million dollars to develop knocked off [hundreds of billions of dollars in NASDAQ market value](#), primarily in AI related companies. The reality is that DeepSeek has a [billion dollar cluster](#) that they used to train their models. And they likely used OpenAI’s ChatGPT [outputs](#) as training data for their models. Nevertheless, you can find DeepSeek’s open-source models available to download for free, no subscription required.

What is reasoning? The models that “reason” use various techniques to accomplish this task, such as a technique named “chain-of-thought”. The models are given time to “think” internally before “speaking”, to produce higher quality outputs, as recently reported in the [Washington Post](#). While this approach can improve AI accuracy and user trust, critics warn that it risks misleading users by anthropomorphizing AI reasoning, potentially prioritizing user appeal over “true” logical rigor.

Remember to show your work! The technique “[chain-of-thought](#)”, is when the model generates intermediate steps that mimic a logical train of thought before arriving at a final answer. This approach allows the model to break down complex problems into manageable parts, leading to more accurate and coherent responses. For instance, when solving a mathematical problem, the model will outline each step of the calculation process,

ensuring that each part is correctly addressed before reaching the solution.

To further refine their reasoning processes, these models are trained using [reinforcement learning](#). In this context, the model is treated as an agent that takes actions (generates responses) based on given states (prompts or questions). The model receives feedback through rewards or penalties depending on the correctness and quality of its responses. Over time, this training method enables the model to develop more effective strategies for problem-solving by learning from its successes and mistakes.

Say what? A reinforcement learning model is like a chef learning a new dish, each step of the recipe is taste tested, and the chef receives praise for the “right” flavors and results, and makes adjustments from negative feedback from adding too much salt. Over time, via trial and error with varying feedback, the chef refines the recipe to create the most satisfying meal.

Models like OpenAI’s o1 are designed to allocate additional computational resources to deliberate on tasks that require step-by-step logical reasoning. This means they spend more time “thinking” before generating an answer, improving their performance on coding, mathematics, and scientific problem-solving. The models can produce more accurate and reliable outputs by dedicating more processing power to these complex tasks.

Models that utilize reasoning techniques are impressive and will likely become ever more so as new algorithms are implemented.

The Anatomy of an o1 Prompt

I want a list of the best medium-length hikes within two hours of San Francisco.

Each hike should provide a cool and unique adventure, and be lesser known.

For each hike, return the name of the hike as I'd find it on AllTrails, then provide the starting address of the hike, the ending address of the hike, distance, drive time, hike duration, and what makes it a cool and unique adventure.

Return the top 3.

Be careful to make sure that the name of trail is correct, that it actually exists, and that the time is correct.

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For context: my girlfriend and i hike a ton! we've done pretty much all of the local SF hikes, whether that's presidio or golden gate park. we definitely want to get out of town -- we did mount tam pretty recently, the whole thing from the beginning of the stairs to stinson - it was really long and we are definitely in the mood for something different this weekend! ocean views would still be nice. we love delicious food. one thing i loved about the mt tam hike is that it ends with a celebration (Arriving in town to breakfast!) The old missile silos and stuff near Discovery point is cool but I've just done that hike probably 20x at this point. We won't be seeing each other for a few weeks (she has to stay in LA for work) so the uniqueness here really counts.

Goal

Return Format

Warnings

Context Dump

Image courtesy of swyx .

The image above is a template for prompting a reasoning model recommended by OpenAI's president Greg Brockman. Notably different from a typical back-and-forth chat with a LLM.

The HUMAN Digest welcomes feedback at kee@lakeforest.edu

Justin Kee, editor-in-chief

Davis Schneiderman, Principal Investigator, [HUMAN](#) grant

AI in higher education

1. Historian Benjamin Breen at UC Santa Cruz, makes the case that while tools like ChatGPT are becoming increasingly disruptive to student learning outcomes, they can be used for [*positive outcomes as well*](#). He describes three use cases.
2. Stanford's Micael Rau, assistant professor of directing and devising in the Theater and Performance Studies, combines [*human creativity and AI*](#) to add to storytelling and stagecraft. His work encompasses image generation (to showcase a dream, for example), large language learning models (to generate a script based on a suggestion) and pose analysis (to create 3D images of actor's bodies for each frame of a recorded performance). According to Rau, AI isn't as superficial as we make it out to be and he believes it can be used to help create meaningful art. Rau's next project is a collaboration with playwright Michael Yates Crowley set to be called [*Hamlet.AI*](#).
3. A Virginia Tech study found that generative AI tools such as ChatGPT, Google's Gemini, and Microsoft's Copilot are gaining traction in higher education. This study provides [*empirical evidence about how VT students and staff use AI*](#), thus allowing for the creation of more informed policies at the university level. One surprising finding was that generative AI usage was incredibly low (70 percent reported using AI less

than once a week) among students and staff. Another significant finding was the difference in usage between STEM and non-STEM students. Students use generative AI easier and more enjoyable than faculty, and STEM students are more positive about its utility than non-STEM students. Male users are also more frequent. Students are of the opinion that AI can aid learning, while the faculty is concerned about students over-relying on it.

General news

1. Researcher Dan Hendrycks, Director of the [Center for AI Safety](#), released “Humanity’s Last Exam”, a dataset of over 3,000 highly challenging questions in over 100 domains of knowledge. Keven Roose over at the NYT authored an [informative piece](#) on the exam on January 23rd. He recently commented on the [pace of acceleration](#) by frontier labs on solving these problems.
2. OpenAI announced [deep research](#), an “o3” (!) reasoning model trained using end-to-end reinforcement learning to navigate complex browsing and reasoning tasks across multiple domains. It can plan multi-step searches, backtrack when needed, browse user-uploaded files, generate and iterate on graphs, and embed relevant images or citations in its responses. This training enables it to achieve state-of-the-art performance on real-world benchmarks.
3. Released to generally underwhelming response, OpenAI also released [Operator](#). Operator is a research preview of an AI agent, which integrates GPT-4o’s vision capabilities with

reinforcement learning to “take control of your computer” and complete multi-step tasks. What could go wrong?

4. The Federal government has changed course on AI to [secure U.S. AI leadership](#) as a matter of national security. The executive order aims to reduce regulations on AI and create an AI “action plan” to maintain U.S. leadership in the field. Additionally, the Stargate Project was announced as a public-private partnership between the U.S. government and private sector AI companies with a goal of building out \$500 billion in datacenters and energy sources over the next four years.
5. The [Vatican](#) weighs in on AI. First, AI should help make our lives better and support human dignity and not try to take over human decision-making or make our moral choices. The Vatican is concerned that we need to be careful that AI doesn't create bigger gaps between people or make us feel less human. And the Vatican makes a really important point: at the end of the day, humans are the ones responsible for moral decisions, not AI. And while AI can do lots of great things, it can't replace real human connections or our search for meaning and truth in life.

Public opinion

A Rutgers University-New Brunswick [survey](#) illuminates American's attitudes towards AI. Overall respondent's attitudes are mixed due to a broad gap in knowledge about AI.

A useful idea

Gradient Descent – This is the fundamental optimization algorithm used in machine learning to adjust a model's parameters (like weights in a neural network) to minimize error. Think of it as hiking down a mountain in the fog: at each step, you move in the direction that lowers your altitude the most, gradually finding the lowest point (the best model fit).

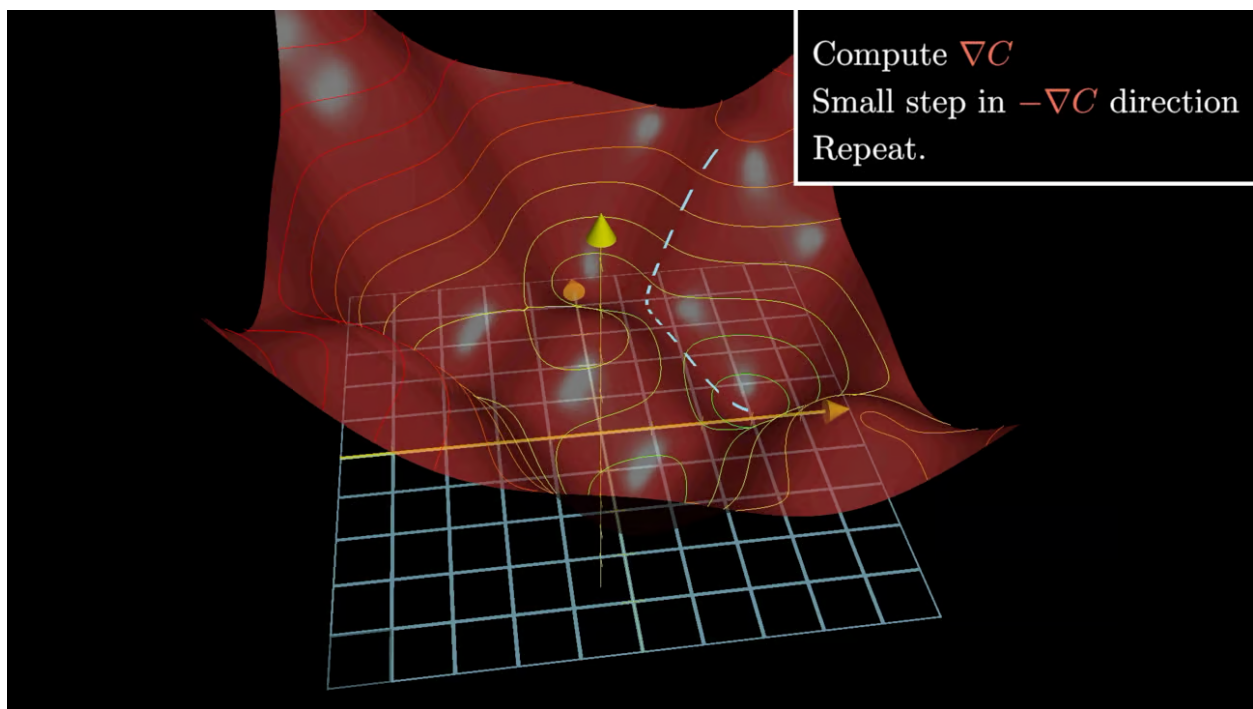


Image via 3Blue1Brown.

A person to know

Yoshua Bengio, born in Paris in 1964, is a renowned computer scientist recognized for his pioneering contributions to artificial intelligence, particularly in deep learning. After completing his PhD at McGill University, he conducted postdoctoral research at MIT and AT&T Bell Labs. Since 1993, Bengio has been a faculty

member at the Université de Montréal, where he founded and serves as the Scientific Director of Mila – Quebec AI Institute, one of the world's leading centers for deep learning research.

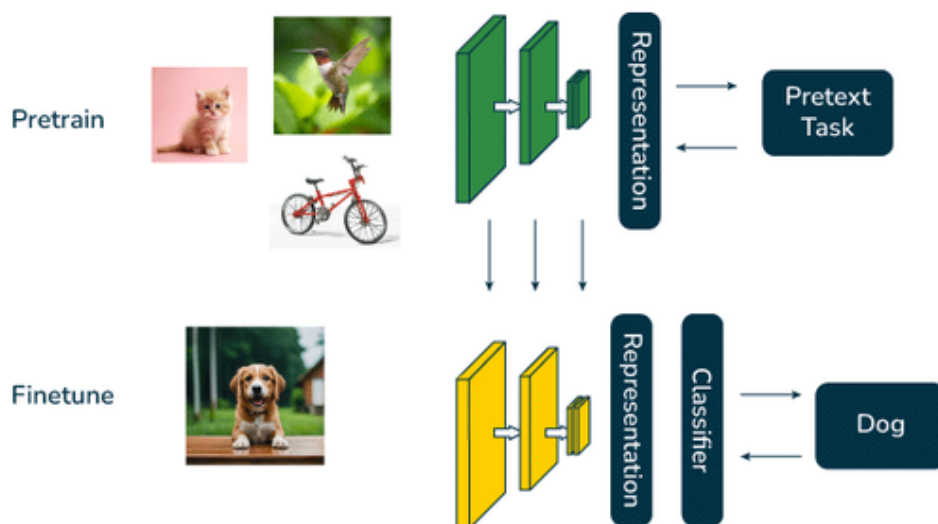


He was awarded the 2018 A.M. Turing Award alongside Geoffrey Hinton and Yann LeCun. His research has been instrumental in developing algorithms and models applicable in areas ranging from speech recognition to autonomous vehicles.

In 2023, he led the [*International Scientific Report on the Safety of Advanced AI*](#), a comprehensive study regarding the safety of current and anticipated general-purpose AI systems. Published in January 2025, the report addresses a broad range of AI related risks.

Technical AI

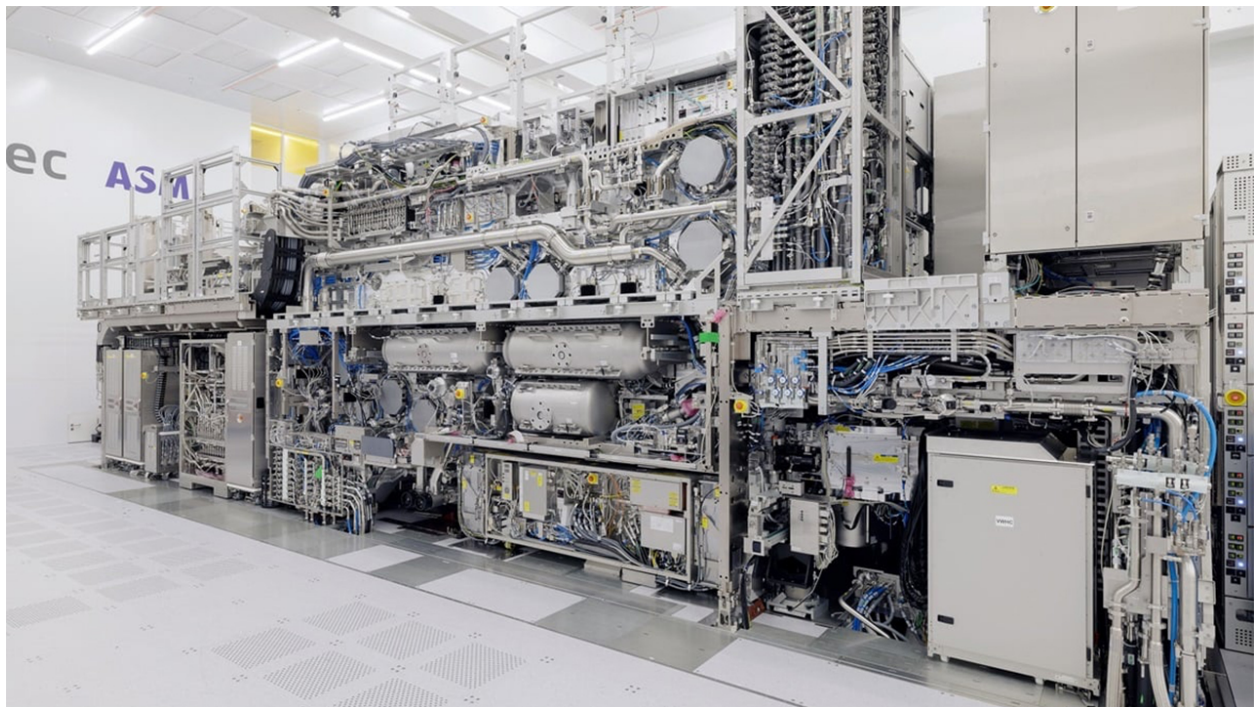
Self-supervised learning. Think about how traditional machine learning works - it usually needs humans to label tons of data, like tagging photos as "cat" or "dog" (See [ImageNet](#)). But self-supervised learning is different as the training data does not need to be manually labeled. It's like giving the AI system a puzzle to solve using raw, unlabeled data. The system creates its own learning signals by doing tasks like predicting missing words in sentences or figuring out what part of an image is missing. This method of training a model is much more efficient. Once an AI system learns the basic patterns from lots of unlabeled data, it only needs a small amount of labeled data to get really good at specific tasks. It's kind of like how a child learns to understand the world by observing and exploring before being formally taught anything.



Something to see

Inside an [ASML machine](#).

High-resolution physics deployed at scale. Advanced Semiconductor Materials Lithography's (ASML) extreme ultraviolet (EUV) lithography machines are at the core of modern semiconductor fabrication, enabling the production of the most [advanced 3nm chips](#).



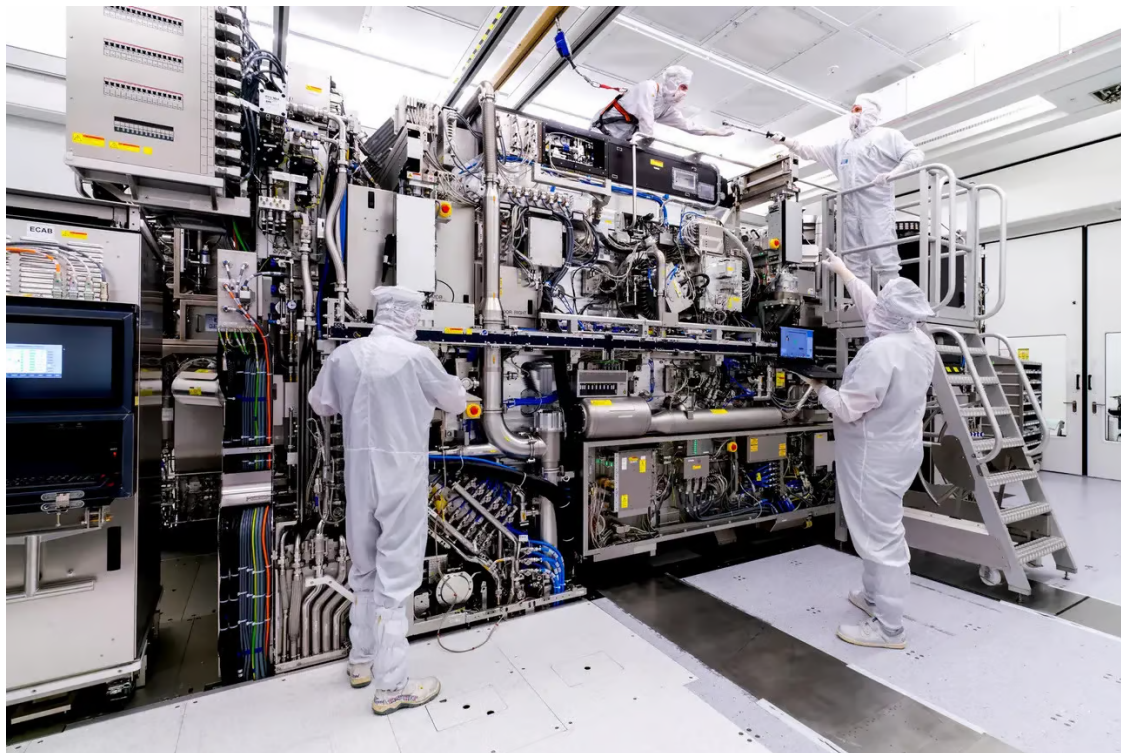
Images via ASML.

The devices are massive, weighing over 70 tons, and expensive, at about \$380 million each. A single machine can produce roughly 16,000 Nvidia GPU chips an hour.

The process of creating an advanced chip begins with a high-purity silicon wafer, which serves as the substrate for integrated

circuits. This wafer is coated with a thin layer of photoresist, a light-sensitive material that reacts when exposed to EUV light. ASML's EUV lithography system projects a meticulously designed circuit pattern onto the wafer using 13.5nm wavelength EUV light, a significant improvement over traditional deep ultraviolet (DUV) lithography. The shorter wavelength allows for much finer feature resolution, making it possible to print transistors and interconnects at the nanoscale, which is critical for achieving higher transistor density.

Unlike conventional light sources, EUV light is produced by firing high-energy laser pulses at tiny droplets of tin, creating a plasma that emits extreme ultraviolet radiation. This light is then collected and precisely guided through a series of specialized mirrors, as EUV wavelengths are absorbed by traditional glass lenses. The light passes through a photomask containing the chip's design and is then projected onto the wafer, transferring intricate circuit patterns with unparalleled accuracy.



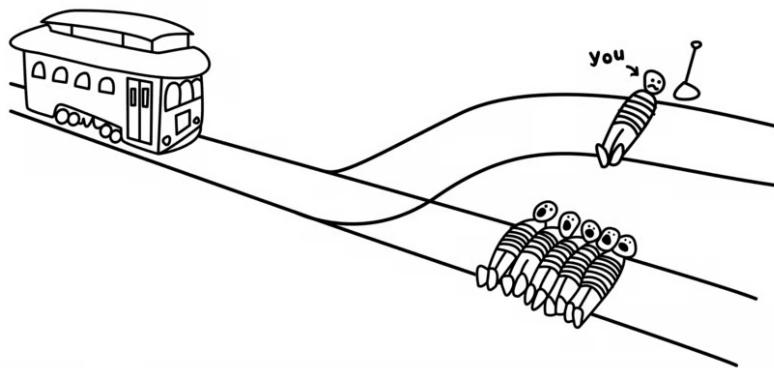
Just for fun

Driving from Los Angeles to San Diego with no [human intervention](#). Sitting in the passenger seat is a flex, for sure. And hopefully there are no [absurd trolley](#) problems encountered along the way.

NEAL.FUN

Absurd Trolley Problems

Level 4: You



Oh no! A trolley is heading towards 5 people. You can pull the lever to divert it to the other track, sacrificing yourself instead. What do you do?

Pull the lever

Do nothing

HUMANS IN THE LOOP.

Thanks to Estelle Ghinea for her research for this issue. Have a tip or suggestion for an upcoming issue of the HUMAN Digest? Write Justin Kee an email at kee@lakeforest.edu.

